Supplementary Material for "ozmV: Per-Sample UTC Time With Declared Uncertainty in an Open Synchrophasor and Waveform Instrument"

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TABLE S-I
PASSES OF THE TWO BENCH SESSIONS (UTC)

Day	Time	Pass	PPS on the 8558A
1	08:23–08:43	M3, slave lag, 10 V and 1000 V, twice	B
1	08:55–09:36	M2-A, voltage delay of A, twice	A
1	11:02–11:06	M3 with the trigger output on the aperture start	A
1	11:14–11:33	M2-A, same	A
1	11:38–11:57	M2-B, voltage delay of B	B
1	12:06–12:28	M3, both ranges, after M2	B
1	07:50–12:49	M1 on both, and A against B, throughout	
2	08:02–10:21	M1 on both; branches swapped at 09:49	
2	09:51–10:21	M4-r02: A against B, nothing else on the PPS lines	
2	10:53–11:33	E4-A: A loses its signal for 1, 5 and 10 min	
2	11:38–12:42	E4-B: B loses its signal for 1, 5 and 10 min	

TABLE S-II
M3: LAG OF THE 8558A BEHIND THE 8588A, BOTH DIGITIZING THE SAME PPS

Pass	Range	Trigger	Edges	Lag (ns)	u (ns)
m3-10V-antes	10 V	ACO	53	49 124.5	21.0
m3-10V-antes-r02	10 V	ACO	59	49 183.1	14.7
m3-1000V-antes	1000 V	ACO	57	49 329.8	32.2
m3-1000V-antes-r02	1000 V	ACO	54	49 322.2	42.4
m3-10V-ape	10 V	APE	58	124.1	12.9
m3-1000V-ape	1000 V	APE	52	315.4	38.2
m3-10V-despues-ape	10 V	APE	57	112.3	13.3
m3-1000V-despues-ape	1000 V	APE	552	326.4	8.9

***Abstract*—This document details the measurements that the paper summarizes: the bench and its passes (S-I), the determination of the converter's delay and the checks of its fit (S-II), the two instruments measuring one signal with independent receivers (S-III), the effect of the time-stamping method on real data (S-IV), the P-class tests, the reporting latency and the cost (S-V), the holdover through losses of the GNSS signal (S-VI) and the start-up window (S-VII). The data behind every table and figure, and the programs that produce them, are in the repository that accompanies the paper. Section numbers of the paper are cited as Section II, III, etc.; those of this document as S-I to S-VII.**

S-I. THE BENCH AND ITS PASSES

Two prototypes of the instrument, A and B, each with its own GNSS receiver (QUESCAN M10TTL, u-blox UBX-M10050) and antenna, measured the same phase of the laboratory mains through their phase-A voltage channel. Each receiver's PPS went to the real-time unit of its instrument and, through a resistive divider, to the neutral-current input of its converter (connector CN5, input AIN0 of the ADS131M08), which the waveform stream recorded in converter volts at 31 502 samples/s together with its per-sample time track. The measurements were made on two days (2026-10-27 and 2026-10-29) at the laboratory of one of the authors; Table S-I lists the passes.

A. The Reference Digitizers

A Fluke 8588A digitized the mains voltage (range 1000 V) and a Fluke 8558A, triggered by the 8588A through the trigger cable, digitized the PPS (range 10 V); both in digitize mode with a sampling interval of 50 µs, an aperture of 49 µs and the input filter off, in bursts of 200 000 readings (10 s), 60 bursts per M2 pass and 6 per M3 pass. The host computer kept time by NTP, which only identifies the UTC second of each PPS edge. For M3 the same PPS went to both instruments through short leads with series resistors.

The trigger output of the 8588A has two modes. Its default, a pulse at the end of each acquisition (`ROUT:TOUT ACO`), starts the 8558A one sample interval after the 8588A: the slave then lags by 49.3 µs, one aperture plus the residual lag, and an analysis that pairs edges within half a sample interval pairs none. With the output as a square wave active during the acquisition (`APE`, falling edge to the slave's external trigger), both instruments acquire together. Table S-II gives every M3 pass: the residual lag is 112 to 124 ns with the master on 10 V and 315 to 326 ns on 1000 V, the master's 1000 V range adding about 200 ns. The M2 analysis uses 325.8 ns with APE (the weighted mean on 1000 V) and 49 327.0 ns with ACO. The per-edge scatter, 100 to 310 ns RMS, exceeds the 82 ns of two independent 200 ns cells, and is larger on the 1000 V range, where the 2.3 V pulse is a small fraction of the range.

B. The Measurement of the Channel's Delay (M2)

A reading of the aperture is the mean of 246 samples taken 200 ns apart, so the reading that straddles a PPS edge places the edge within one 200 ns cell. The edges of successive seconds fall at different phases of those cells, because the digitizers' time base is not locked to the PPS, so the mean of many edges is unbiased and its standard error falls as $1/\sqrt{\text{edges}}$ from the 58 ns RMS of one cell; the ten edges of a burst fix the time base of that burst to about 20 ns, and an edge that falls in the 1 µs gap between apertures is dropped. The phase of the 50 Hz voltage against UTC, from the 8588A's readings on that time base (minus the slave's lag, plus half the aperture) and from the instrument's samples with

TABLE S-III
M2: DELAY OF THE INSTRUMENT'S PHASE-A VOLTAGE LABELS BEHIND THE 8588A'S PPS-REFERENCED TIME BASE

Pass	Instr.	Trigger	Windows	Delay (µs)	Windows RMS (ns)	u (ns)
m2-A-VA	A	ACO	2895	72.159	47	11.2
m2-A-VA-r02	A	ACO	2768	72.174	47	12.9
m2-A-VA-ape	A	APE	2940	72.225	47	4.9
m2-B-VA-ape	B	APE	2882	72.713	94	8.0

TABLE S-IV
M1: OFFSET OF THE SAMPLE LABELS AGAINST THE INPUT, BEYOND THE GROUP DELAY OF 190.5 t_{MOD}

Instr.	Day	Recordings	Edges	Offset (ns)	Spread (ns)
A	1	10	16 493	11 876.9	20.2
B	1	9	15 240	11 899.5	28.6
A	2	4	7103	11 867.9	30.6
B	2	4	7137	11 830.0	33.5

their own labels, compared in windows of 0.2 s by a four-parameter sine fit, gives the delay by which the instrument's labels lag the input. Table S-III gives the four passes. The delay so measured is the phase delay of the channel at 50 Hz; at other frequencies it differs wherever the front end's phase is not linear, and the waveform stream, which also needs it, is outside this measurement. An oscilloscope with both probes grounded at the PPS node saw no difference between the edge at the instrument's PPS wire and at the 8558A's input at 100–200 ns per division; the check at the processor's own pin was not possible, the pin being hidden under the module.

C. Loads on the PPS Line

The receiver's pulse was shared, at times, with the 8558A and with the oscilloscope's probes. Whenever an instrument hung on one of the two PPS lines, the phase difference between the two instruments' phasors (S-III) moved by 100 to 300 ns: it stepped by −0.27 µs at 11:35 of the first day, when the 8558A moved from A's pulse to B's, and by about 0.3 µs at 08:36 of the second day, when the oscilloscope's probes were connected to A's. A load on the line delays the edge that the instrument's capture input sees. The reference digitizer on the same node sees the same delayed edge, so the delays of Table S-III are unaffected; the time against UTC is, and the receiver's specification holds for a pulse that drives the instrument's input alone. The faults of the prototype boards' acquisition while cables were being changed, and the reboots they caused, are matters of the prototypes and do not enter the measurements.

S-II. THE CONVERTER'S DELAY

A. Simulation of the Measurement

It may seem surprising that a 31.5 kS/s converter can locate an edge to nanoseconds. Each output sample, however, is a weighted average of 382 modulator samples taken every $t_{\mathrm{MOD}} = 248$ ns, so the amplitudes of the samples around one edge locate it to within about 72 ns RMS, the quantization of the modulator grid, and the edges of successive seconds fall at different phases of that grid. A simulation of the measurement (`analysis/m1_pps_step_sim.py`), with the converter's modulator and sinc^3 filter, the noise of its datasheet (21.3 µV RMS at this gain and rate) and the measured errors of the sample times (28 ns RMS per second from the tick-to-UTC model, 2.8 ns per sample from the envelope regression), recovers an offset of $44\,t_{\mathrm{MOD}}$ to 2 ns RMS from 600 edges with a sharp edge at the input, and to 6 to 13 ns when the RC network must be fitted as well. The precision comes from the fit of hundreds of edges, not from the spacing of the observations; the delay and the time constant of the network are strongly correlated in the fit, so the model of the network matters more than the noise.

B. The Measured Offset, Recording by Recording

Every 30-minute recording of both days with at least 1000 usable edges and the instrument's own pulse on its input was fitted on its own (`analysis/s01_wmu_batch.py`, which runs `analysis/m1_pps_delay.py` on each). Table S-IV gives the means per instrument and day; the recordings of one day scatter by 20 to 34 ns about them. The pulse was 338 mV at the converter input, the samples' noise 0.2 mV; in one recording the 1733 edges fell on 3930 distinct phases of the sample period, never more than 175 ns apart. The time constant of the one-pole input network comes out at 32.8 (A) and 32.7 µs (B), against about 27 µs from the nominal series resistors and capacitor of the input: the difference is the source resistance of the divider that feeds the pulse, which the fit absorbs. The model-free centroid of the equivalent-time impulse response, delay plus time constant, is 44.73 (A) and 44.64 µs (B).

C. Checks of the Fit

On one recording (instrument B, first day, 08:48–09:18 UTC, 1733 edges; `analysis/m1_models.py` and `analysis/m1_windows.py`):

- the level windows of the fit, which define the low and high states of the pulse, move the offset between 11 867 and 11 932 ns (the paper's windows give 11 879), and the two halves of the recording give 11 896 and 11 862 ns;
- a second pole added to the input network collapses to 1 ns and leaves the offset at 11 878 ns;
- a direct feed-through added to the one-pole network comes out at $-0.9\,\%$ of the step, unphysical for a passive network, with the offset at 11 594 ns; the one-pole model is kept.

The residual of the fit is 6.4×10^{-4} of the step RMS; Fig. S1 shows it in bins of 2 µs for both converters. From these checks the paper takes $u_D = 40$ ns.

D. The Time Base Checked Against the Pulse, Second by Second

The same edges, fitted one second at a time with the time constant fixed at the recording's value, give the time of each edge from the labels of the samples around it.

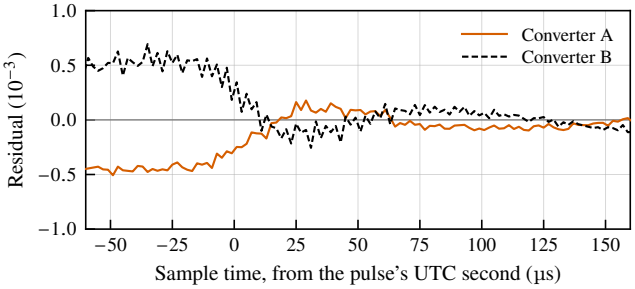


Fig. S1. Residual of the fit of Fig. 2 of the paper, for each converter, in bins of 2 µs.

TABLE S-V
B MINUS A: THE TIME BY WHICH B'S PHASOR LAGS A'S, PER PERIOD

Period	Seconds	Mean (ns)	RMS (ns)	5-min m
Day 1, instruments on the PPS lines	15 765	584.8	209	280–
Day 2, 08:12–09:14, same	3477	746.7	182	453–
Day 2, M4-r02, nothing on the lines	1756	504.8	72	473–
Day 2, E4 with the cross PPS cabling	5977	499.7	115	361–765

Over the recording above, that time scatters by 46 ns RMS from one second to the next about its median. On the bare sample grid (sample index times a constant period, which removes the labels and leaves the pulse and the measurement) the scatter is 33 ns, and the difference between the two, the labels' own second-to-second noise, is 15 ns (`analysis/s01_grid_vs_labels.py`). The tick-to-UTC model of Section II-C was chosen on the same basis: on the bench, the two-anchor line predicted the next pulse with an error of 14.6 ns RMS, least-squares lines through 3, 4 and 8 anchors did about as well, 12.2 to 14.9 ns, and one through 16 anchors did much worse, 35.1 ns, because over more than a few seconds the wander of the oscillator's frequency matters more than the jitter of the pulses, about ±10 ns from one pulse to the next.

S-III. TWO INSTRUMENTS ON ONE SIGNAL

The two instruments' phasors were paired by their UTC stamps, never by their arrival, and their waveform streams compared on the same UTC seconds through their time tracks (`analysis/s01_ab_pmu.py`, `analysis/s01_ab_wmu.py`). Table S-V gives the difference, B's phase behind A's expressed as a time, in each period. For the same seconds the two streams of one instrument give the same difference to 2.0 ns RMS (mean 0.0 ns). The amplitude ratio B/A was 0.999 69, the frequency difference 1.0×10^{-4} Hz RMS and the ROCOF difference 0.023 Hz/s RMS. In the clean period, with nothing but the two instruments on the PPS lines and the voltage branches swapped between them (the difference did not move with the swap, so it lies in the instruments' channels and not in the wiring), B lags A by 504.8 ns; the channel delays of Table S-III differ by 487.4 ns, and the residual, 17 ns, is the disagreement of the two receivers and time chains, with the caveat that two receivers of one model may share a bias against UTC.

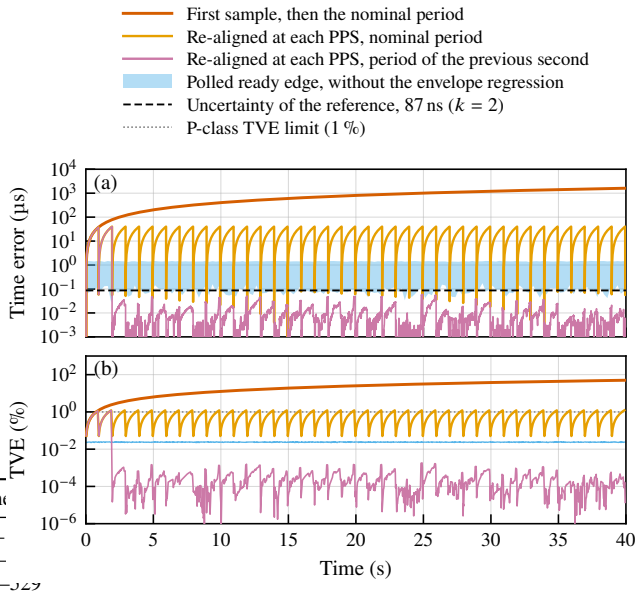


Fig. S2. The same 40 s of mains voltage (1997 reports) time-stamped in four common ways, against the per-sample time of the paper. (a) Time error of each method; for the polled ready edges, the band between the 5th and 95th percentiles. (b) TVE that each method causes in the reports of the production estimator.

S-IV. THE EFFECT OF THE TIME-STAMPING METHOD ON REAL DATA

To isolate the effect of timing, we took 40 s of mains voltage recorded by the instrument together with its per-sample times (release 17 of the software, 2026-10-14), re-timed the same samples, in simulation, in the ways reviewed in Section I of the paper, and ran each version through the same estimator (`analysis/e3_dating_strategies.cpp`; Fig. S2). The errors are differences from the reports computed with the per-sample time, and so measure the effect of the time-stamping alone.

Time-stamping the first sample and extrapolating with the nominal period accumulates the 40 ppm of the crystal: 1.61 ms in 40 s, a TVE of 50 % and an RFE of 0.42 Hz/s. Re-aligning at each PPS limits the error to the drift of one second, 40.3 µs, which is still a TVE of 1.2 % just before each pulse and an RFE of 0.023 Hz/s, and the jump at each pulse breaks the windows that straddle it: 120 of the 1997 reports were lost and the estimator restarted 40 times. With the period of the previous second, the drift almost disappears once a full second has been measured: from its third second on, this method stayed within 59 ns of the per-sample time, less than that time's own uncertainty, and its largest errors and its 6 lost reports fell in the first two seconds. The polled ready edges without the regression of Section II-B, simulated by adding to the per-sample times a lateness drawn uniformly from the measured range of 0 to 1.52 µs, cost only 0.026 % of TVE, but 0.079 Hz/s of RFE, a fifth of the P-class limit, with no other error in the chain. Times truncated to whole microseconds err by up to 1.00 µs and cost 0.016 % of TVE and 0.006 Hz/s of RFE.

S-V. P-Class Tests, Latency and Cost

A. The Two Parts of the Change

The change of the estimator described in Section IV-B of the paper had two parts: the phasor from the first-order sub-model instead of the full second-order model, and frequency and ROCOF from the positive sequence instead of phase A. A factorial run of the four combinations through the P-class tests (`analysis/e5_factorial.sh`, `data/e5/factorial`) separates their effects: the phasor order alone removes the step overshoot and sets the harmonic, modulation and ramp TVE of Table II of the paper, and the reference alone sets the ramp RFE and the noise-limited RFE, each leaving the other's figures untouched. The steps were evaluated as the standard prescribes: equivalent-time sampling over 20 step positions, state levels and $\pm 0.5\,\%$ boundaries for the response time, the delay time and the overshoot (5.2.3, 5.2.4 and Annex C.4 of IEC/IEEE 60255-118-1). Earlier figures of 40 ms of response time and 3.6 ms of delay, obtained before the harness followed the standard's method, were artefacts of that harness.

B. Reporting Latency

The latency was measured on both instruments during the second day's M4-r02 pass, from the C37.118 frames that each instrument published to its own network interface and captured there with their arrival times (89 998 and 89 967 frames): the arrival time minus the report time of each frame is 31 to 52 ms from the 1st to the 99th percentile, with a median of 41.3 (B) and 41.1 ms (A) and maxima of 100 and 63 ms. The P class allows $2/F_s = 40\,\mathrm{ms}$. The half window of the fit is 20 ms and the half length of the decimation filter 4.1 ms; the rest comes from the batches of 256 samples (8.1 ms) in which the acquisition hands samples to the estimator, and from the scheduling of the estimation and the publishing. The arrival times are those of the instrument's system clock, which the time service disciplines to the receiver; this measurement does not verify that clock to better than a few milliseconds.

C. Computational Cost

On one Cortex-A53 core at 1.4 GHz, the estimator takes 3.5 % of the core for three voltage channels at 50 frames/s, decimation filter included, about 0.7 ms per report with up to three carrier refinements; fitting every sample instead of one in 16 would take 19 %. For comparison, a single-channel two-cycle Taylor–Fourier estimate at 8 kHz takes about 7 ms on a BeagleBone Black (Tosato *et al.*, 2018, cited in the paper). The whole measurement service of the instrument, which also computes the power-quality indices, takes 25.5 % of one of the four cores with the PMU off and 35.5 % with it publishing 50 frames/s; of those ten points the estimator accounts for 3.5, and the rest goes to handing every sample, with its time and time quality, to the estimator and to publishing the reports. Streaming every sample with its time track adds another 14 points, so the instrument runs both at about half of one core.

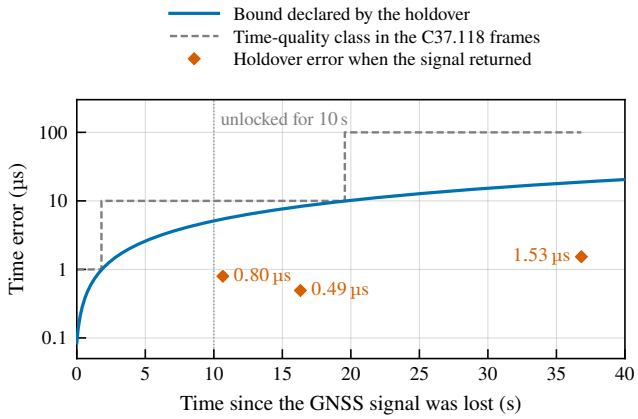


Fig. S3. Holdover through losses of the GNSS signal during one hour on the bench. Solid line: the bound declared from the start of a loss with a fresh period. Dashed staircase: PMU time-quality class carried by the C37.118 reports during the longest loss. Diamonds: distance between the holdover and the GNSS time when the signal returned, for the three losses of the hour.

S-VI. Holdover Through Losses of the Signal

A. One Hour on the Bench

Fig. S3 summarizes one hour on the bench, with the antenna under open sky, during which reception was lost three times, twice because the antenna was covered by hand. The PMU published 179 947 reports at 50 frames/s, one at every instant of the 20 ms grid, and its estimator never restarted. During the losses, which lasted 16.3, 36.8 and 10.7 s, the time-quality code of the reports followed the declared bound through the decades from 100 ns to 100 µs, and the unlocked-time bits changed after 10 s. The receiver came back 493, 1531 and 796 ns away from the holdover, no more than 15 % of the bound declared at each instant. The waveform stream flagged every sample of the three losses, with an uncertainty that grew to 18.7 µs over the longest, and lost none. Before this design, a single loss had cost the PMU 2.96 s of reports and two restarts of its estimator. In an earlier bench run with the antenna in its deliberately poor position, 22 of 23 losses ended within 2.4 µs of the holdover, and the remaining one, after 8 min, ended 15.2 µs off, about 5 % of the bound at that length; under open sky, over eight hours of logging, the receiver returned between 19 and 1324 ns from the holdover, with a median of 112 ns.

B. The Drift Allowance

The drift allowance of the bound, $\delta = 0.5\,\mathrm{ppb/s}$, comes from the bench crystal itself: over ten hours of logging (`analysis/crystal_drift.py`), its frequency changed by up to 0.27 ppb/s averaged over an hour, 0.40 ppb/s over half an hour and, once, during a thermal transient, 0.68 ppb/s over ten minutes. Even that rate, sustained for a whole hour from an exact period, would leave the holdover 4.4 ms off, within the 5.0 ms that the bound declares by then. A return within 10 µs, a third of the sample period, continues the same time scale; a larger one starts a new one, and the waveform record marks the discontinuity as a gap.

TABLE S-VI
E4: ERROR OF THE HOLDOVER AT THE END OF EACH LOSS, AGAINST THE OTHER INSTRUMENT'S PULSE

Loss	Length (s)	Error at end (µs)	Largest (µs)	Bound at end (µs)	Own check (µs)	Drift pred (µs)
A, 1 min	100	1.74	2.04	52.9	1.85	2.1
A, 5 min	349	13.75	13.75	206.1	13.63	10.6
A, 10 min	647	40.39	40.40	430.8	40.52	28.9
B, 1 min	98	3.57	3.57	51.6	3.42	1.4
B, 5 min	331	13.79	14.08	193.8	14.13	6.5
B, 10 min	660	22.05	22.10	440.6	21.97	7.9

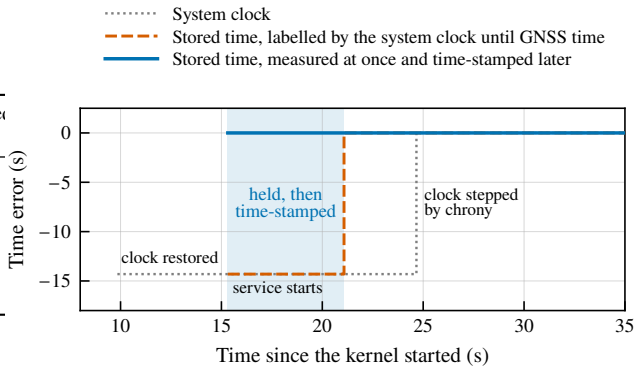


Fig. S4. Start-up after a reboot, a typical boot from the means of 20 (`data/boot-after-r20.csv`): the error of the time stored with each measurement, before the change (the restored clock) and after it (the provisional window, whose records are time-stamped once the GNSS time is known).

C. The Six Losses of 1, 5 and 10 Minutes

Table S-VI gives, for each loss of Fig. 4 of the paper, its length (from the first to the last sample timed by the holdover), the error at its end and the largest error during it against the other instrument's pulse, the bound declared at the end, the instrument's own return check, and the error that a linear frequency drift would have accumulated, computed from the change of the measured period across the loss (`analysis/s02_e4.py`). The error is the offset of the labels against the reference pulse minus the median offset over the five minutes of confirmed GNSS time before the loss, which removes the converter's delay and the receivers' constant difference. The instrument's own check is the distance between the holdover and the first confirmed sample at the return; returns beyond 10 µs start a new time scale, which the waveform track flags, so these are the time steps of the instrument rather than the rejoins it counts within the same scale. The antenna was covered by carrying it indoors into a closed box, which took 20 to 30 s; in one case (A, 10 min) the receiver saw the sky for 4 s during that transit without confirming its time, and the loss is counted from the actual loss of the fix.

The shorter spontaneous holdovers of the two days, 45 of 10 to 29 s on the first day with the instrument's own pulse on its input, gave errors of at most 0.99 µs, never more than half the declared bound apart from one disturbed second while cables were being changed, and the instrument's own return check agreed with the independent measurement to 10 to 70 ns; one natural loss of 5.2 min on the first day, during which the receiver stopped its pulse, ended with the instrument's own check at 38 ns against a bound of 182 µs.

S-VII. MEASURING BEFORE THE TIME IS KNOWN

The bench board has no battery-backed clock. At each boot it restores the time saved at shutdown, and the time service steps the system clock by the length of the outage, about 14 s on the bench, once it selects the receiver, some 25 s after the kernel starts; the measurement service starts earlier, about 15 s after the kernel. Labelled with the restored clock, its first records would be time-stamped 14 s early, and before the change described here they were: in each of 20 reboots, one or two of the power-quality records, made every 3 s, were stored inside the real outage, and the stored data showed an outage 14 s shorter than it was.

The instrument now measures from its first sample and assigns the time afterwards. A session that starts before its time is known opens a provisional window, in which samples are labelled by counting nominal periods from the first one and everything the analyzer would store is held back; the first sample with a confirmed epoch closes the window. Because every conversion is counted, the final time of each held sample follows from the first confirmed sample by counting conversions back at the period measured with GNSS, so no sample is misnumbered and the error is the oscillator's drift over the window, a few nanoseconds; the held records are stored with it. In 20 reboots (Fig. S4) the window lasted 5.3–6.3 s and was always closed by GNSS time; it moved the held records by 14.04–14.38 s, at a rate that corrected the crystal's 40 ppm, no record was stored on the restored clock, and the stored data showed the true outage, 26.2–27.7 s.